

ANATOMY AND PHYSIOLOGY DEFINED



- **Anatomy**

- The study of structure and the relationships among structures.

- **Physiology**

- The study of how body structures function



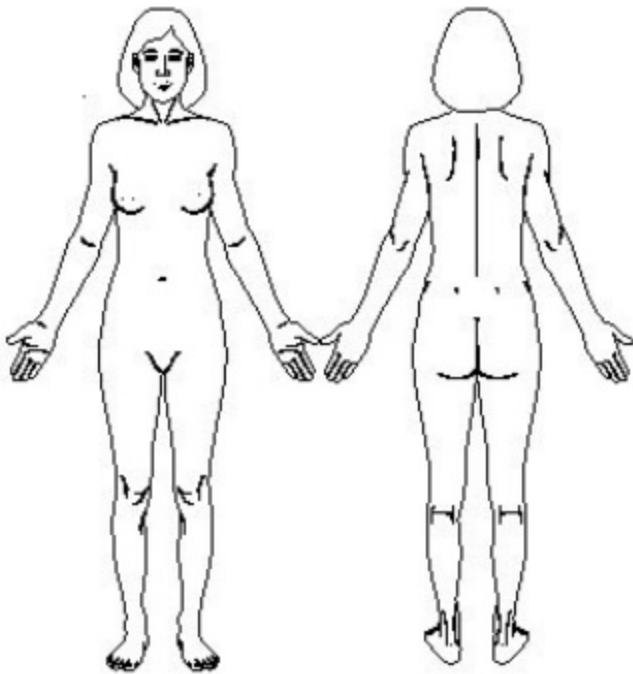
Anatomical position

- Body is erect with the feet parallel and the arms hanging at the sides with the palms facing forward.

Other positions of body



1. Anterior/Posterior



- Anterior—to be situated near or toward the front of the body
- Posterior--to be situated toward the back of the body

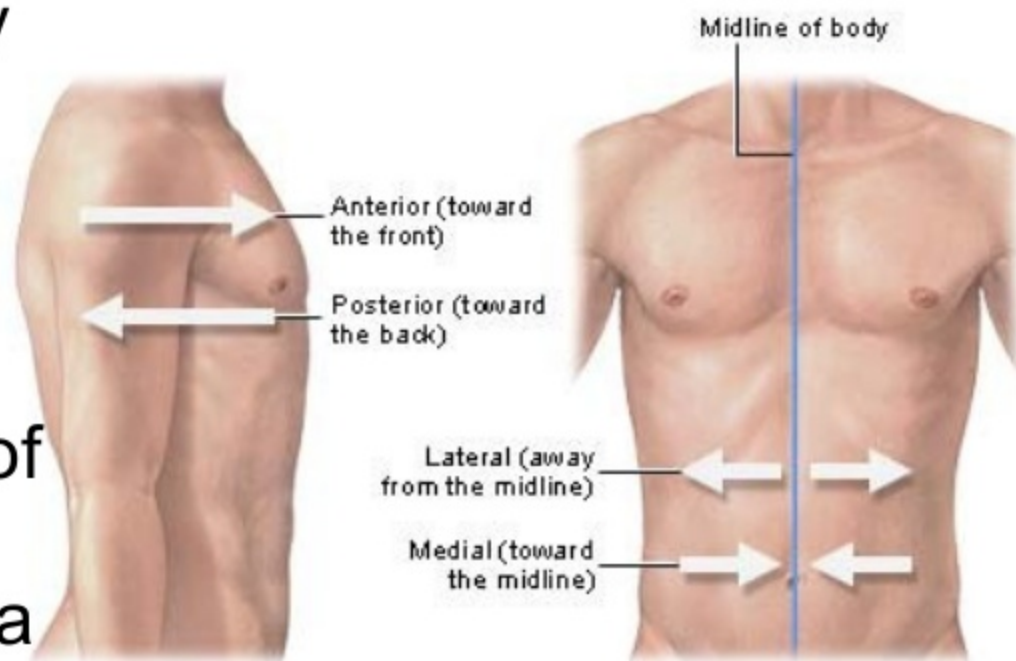
2. Superior/Inferior Cranial/Caudal



- Superior—means to be situated toward the upper part or head of the body, positioned above another organ or structure
- Inferior—to be situated toward the lower part of the body or positioned below another organ or structure
- Cranial—refers to the head end
- Caudal—refers to the tail end

3. Medial/Lateral

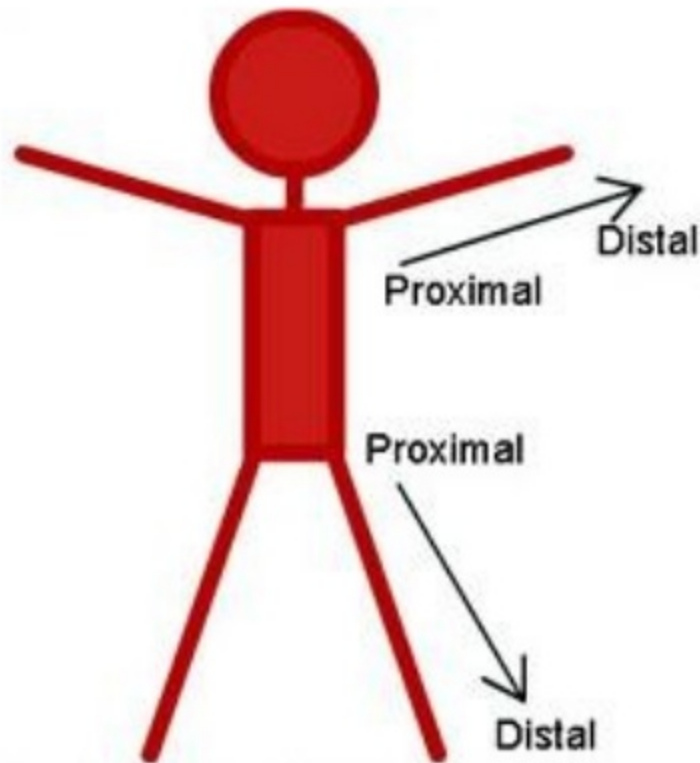
- Medial—to be closer to the midline of the body or a structure, being internal as opposed to external
- Lateral—to be farther away, in the direction of either side, from the midline of the body or a structure

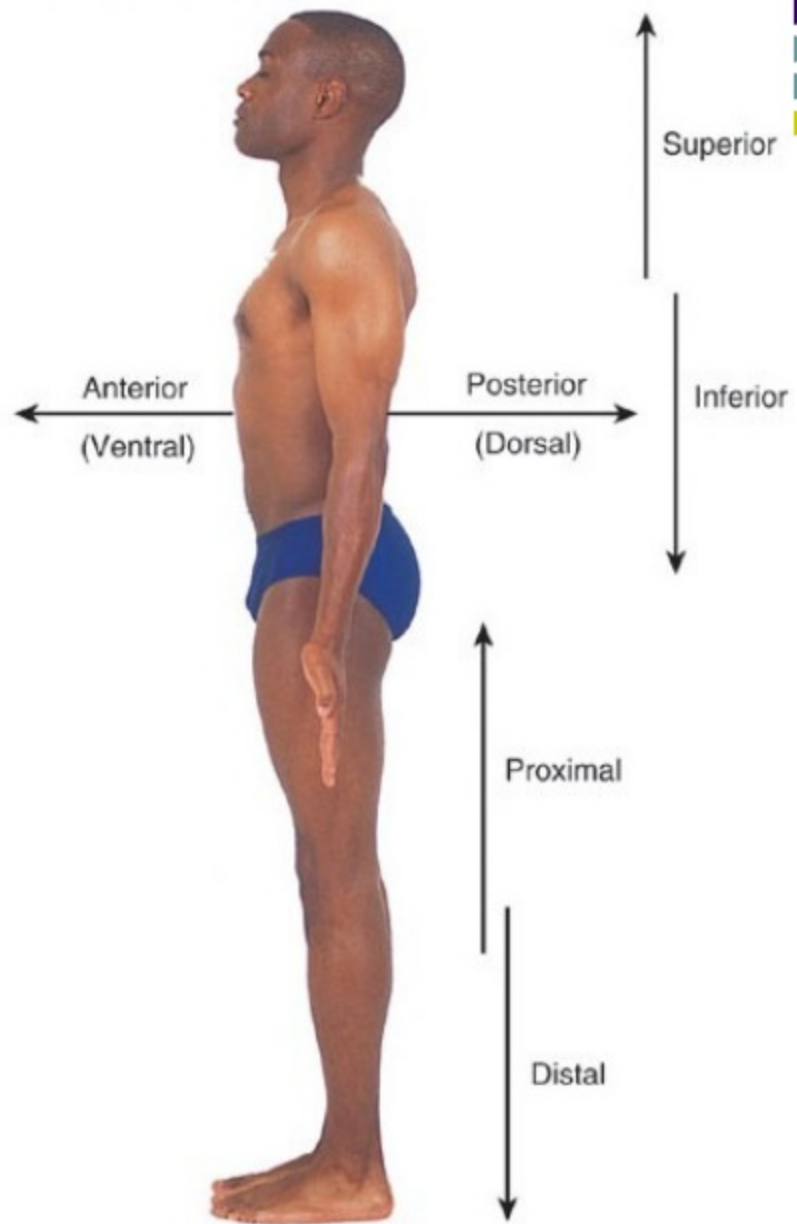
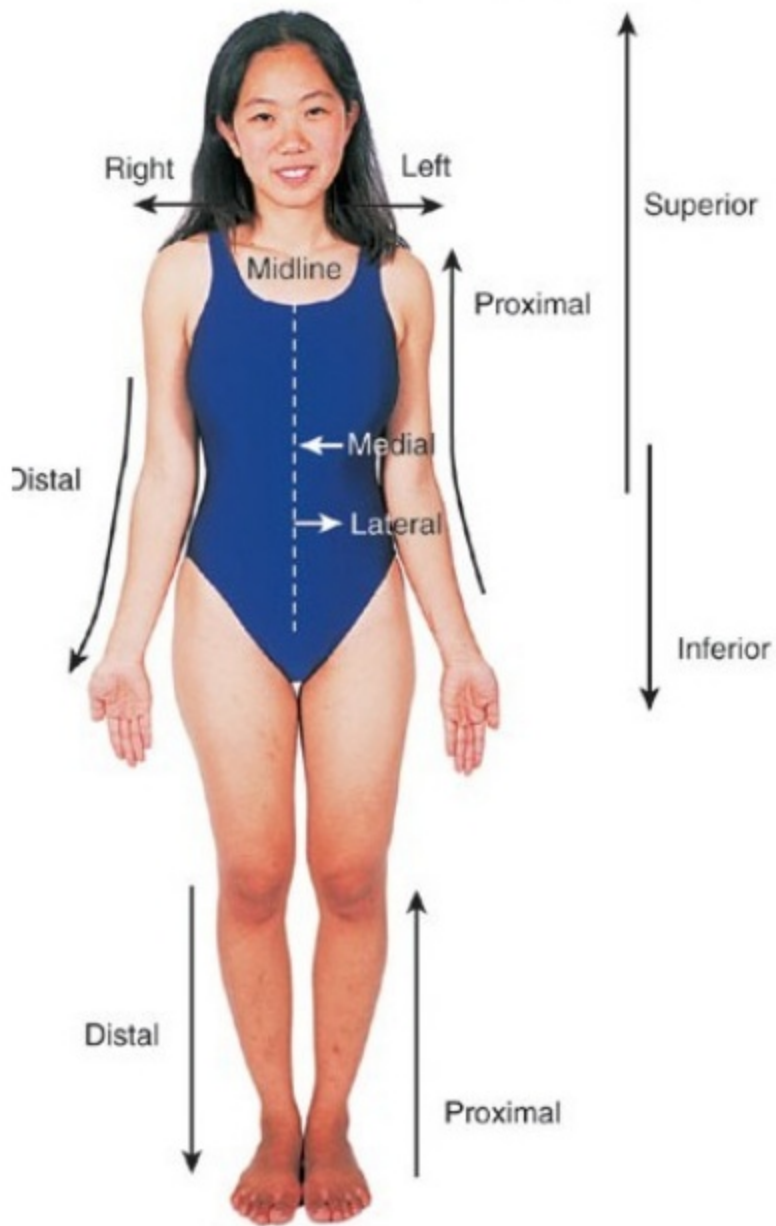


4. Proximal/Distal



- Proximal: Toward or nearest the trunk or the point of origin of a part
- Distal: Away from or farthest from the trunk or the point of origin of a part





5. Unilateral/Bilateral



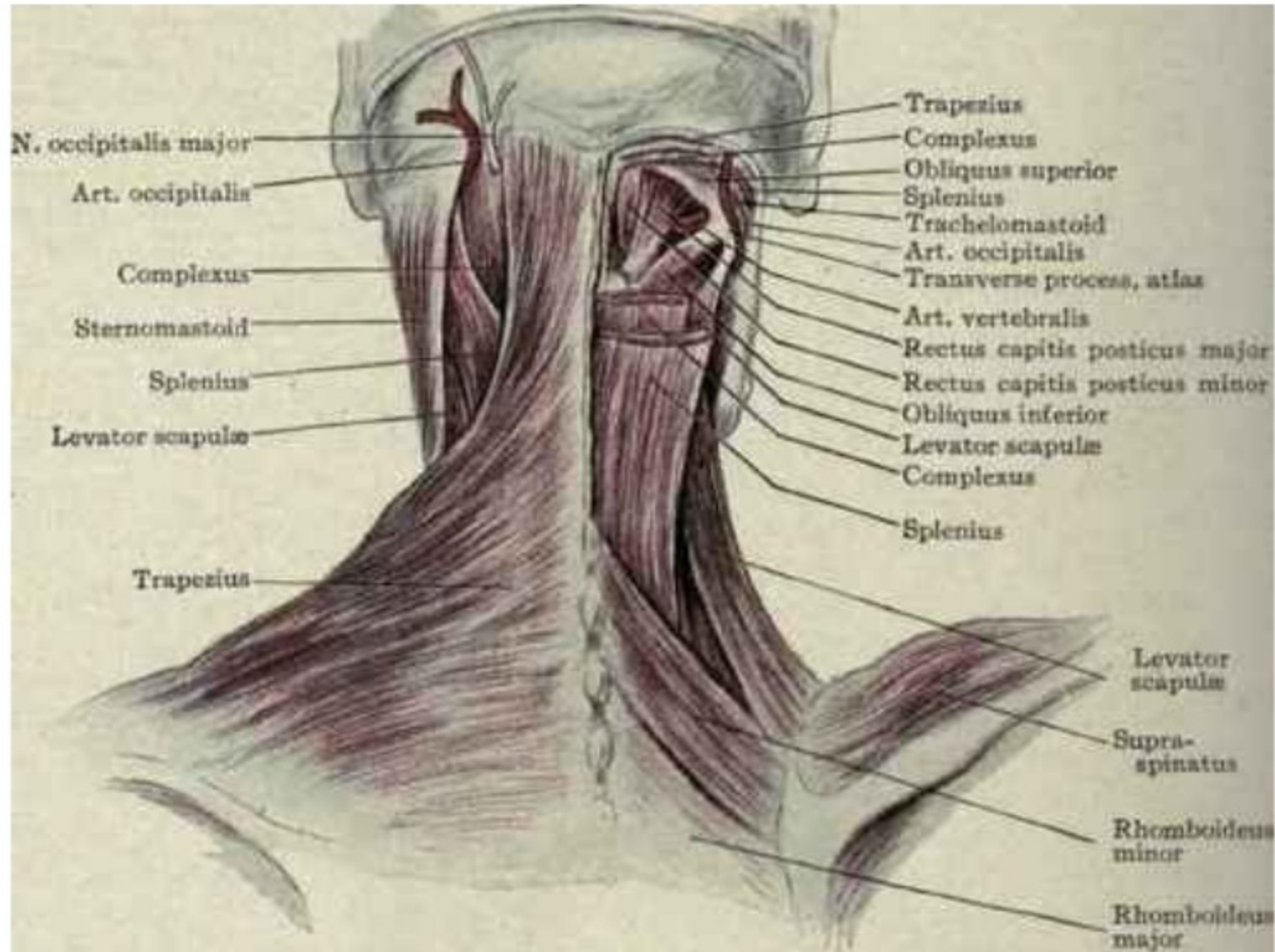
- Unilateral—on one side
- Bilateral—on both sides

6. Superficial/Intermediate/Deep



- Superficial—near the outer surface of the body
 - i.e. skin is **superficial to** the muscle layer
- Intermediate—between two other structures
 - i.e. the naval is **intermediate to** (or **intermediate between**) the left arm and the contralateral (right) leg.
- Deep—further away from the surface of the body
 - i.e. the muscular layer is **deep to** the skin, but superficial to the intestines.

Superficial/Intermediate/Deep





Body planes

These are based on four imaginary planes that pass through the body in the anatomical position. They are

- 1.Sagittal
- 2.Midsagittal (median) plane
- 3.Frontal (coronal) plane
- 4.Transverse plane (cross section)

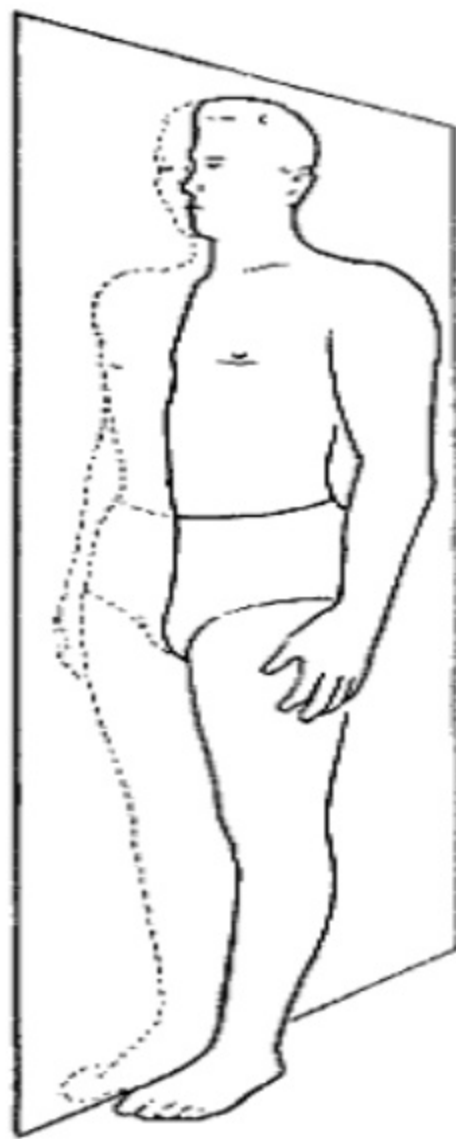


Body planes

1. **Sagittal Plane** – divides body into right and left parts.

2. **Midsagittal/median**

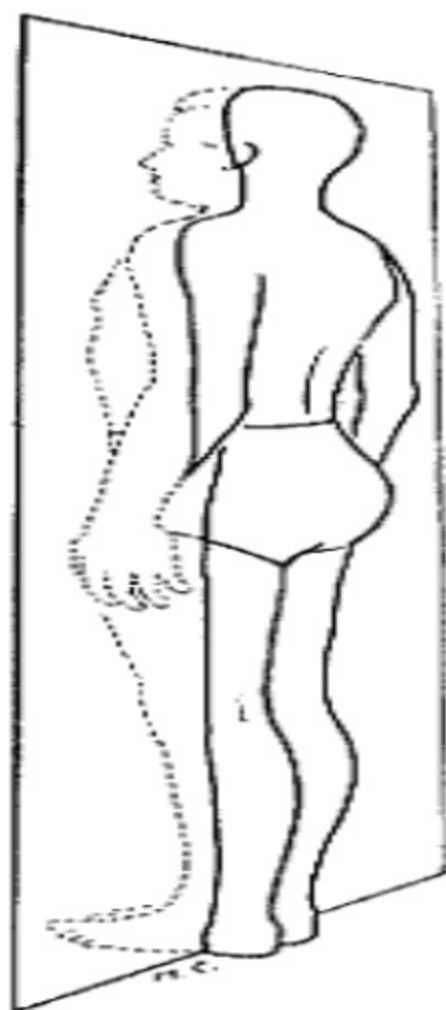
plane –divides body into two equal halves



Sagittal/Anteroposterior
Plane

Body Planes

3. Frontal /coronal plane – divides body into anterior and posterior parts



Frontal/Lateral
Plane





Body Planes

4. Transverse plane/cross

**Section/horizontal
section** divides into
upper and lower parts

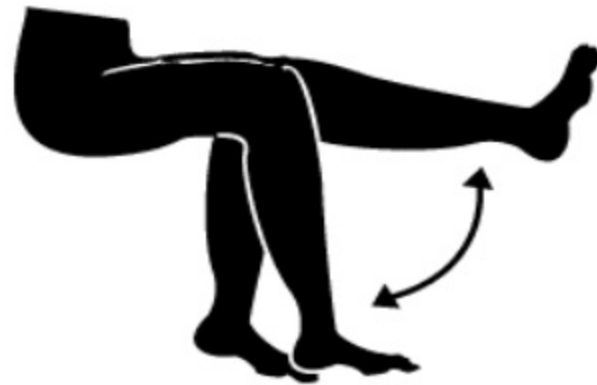


Horizontal/Transverse
Plane

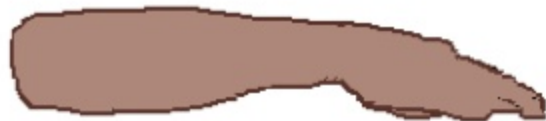
Terms of Movement



1. Flexion/Extension



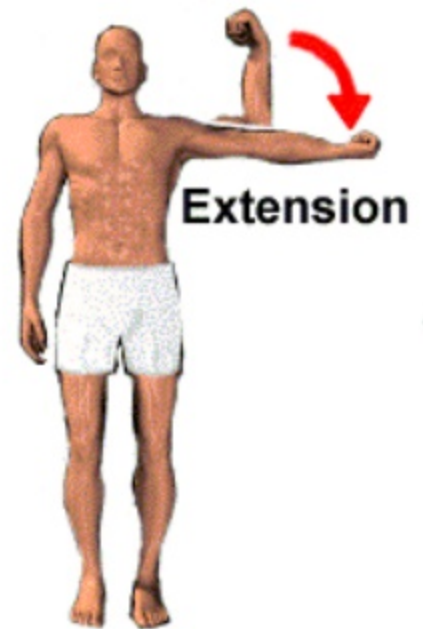
Extend Wrist



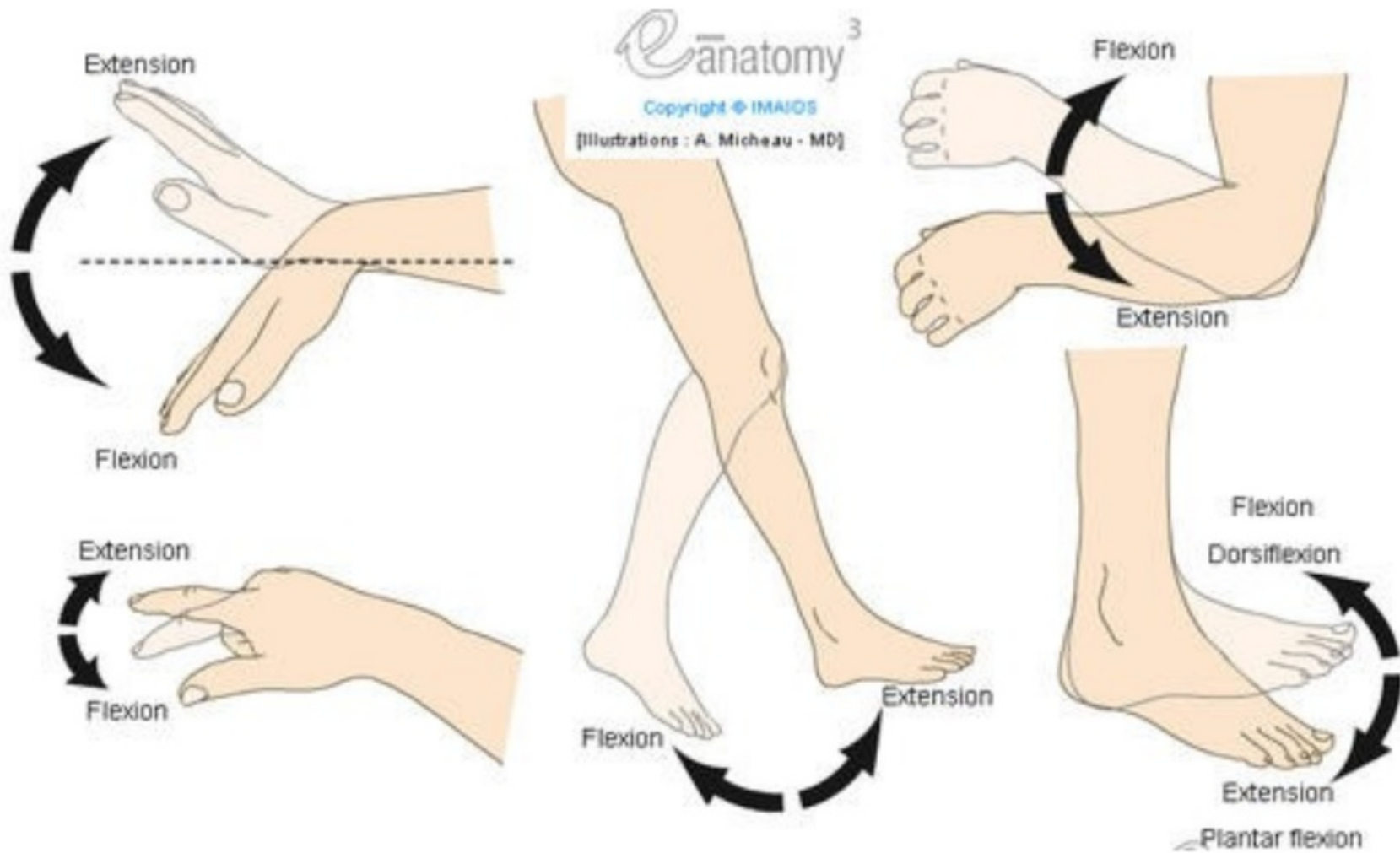
Flex Wrist



Flexion



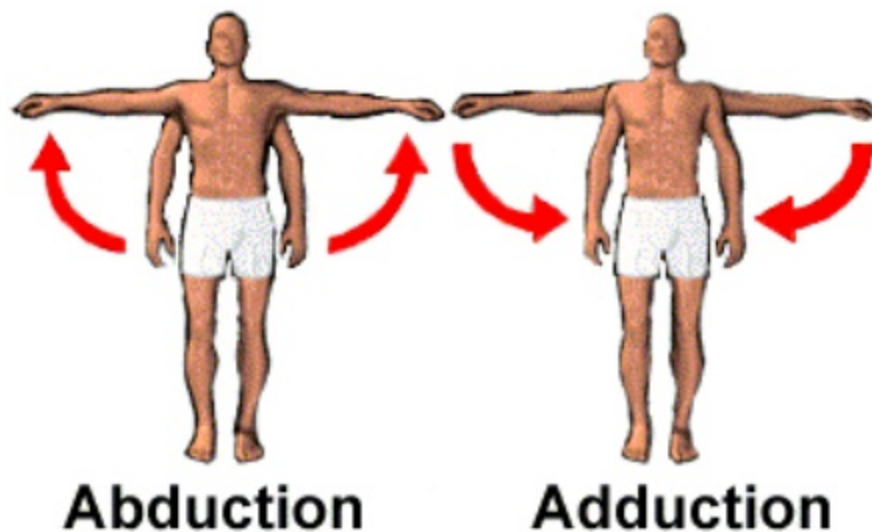
Extension





2. Abduction/Adduction

- Abduction—moving a body part away from midline
- Adduction—moving a body part toward the midline

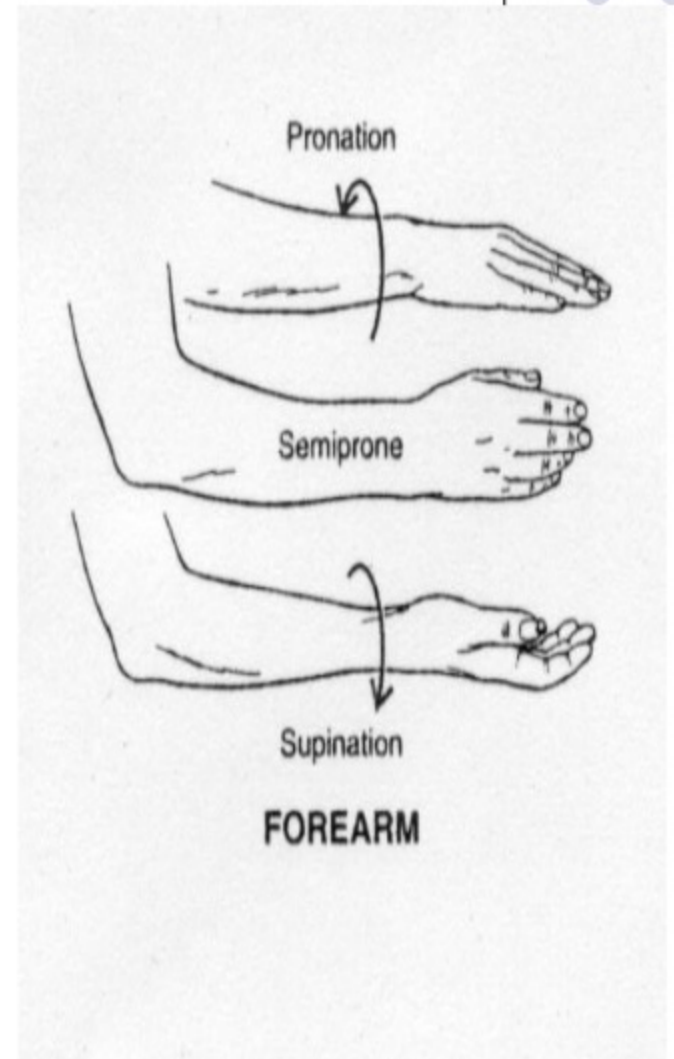




3. Supination/Pronation

❖ Supination- Rotating hand/forearm laterally
“hold the soup”

❖ Pronation- Rotating hand/forearm medially
“pour the soup”

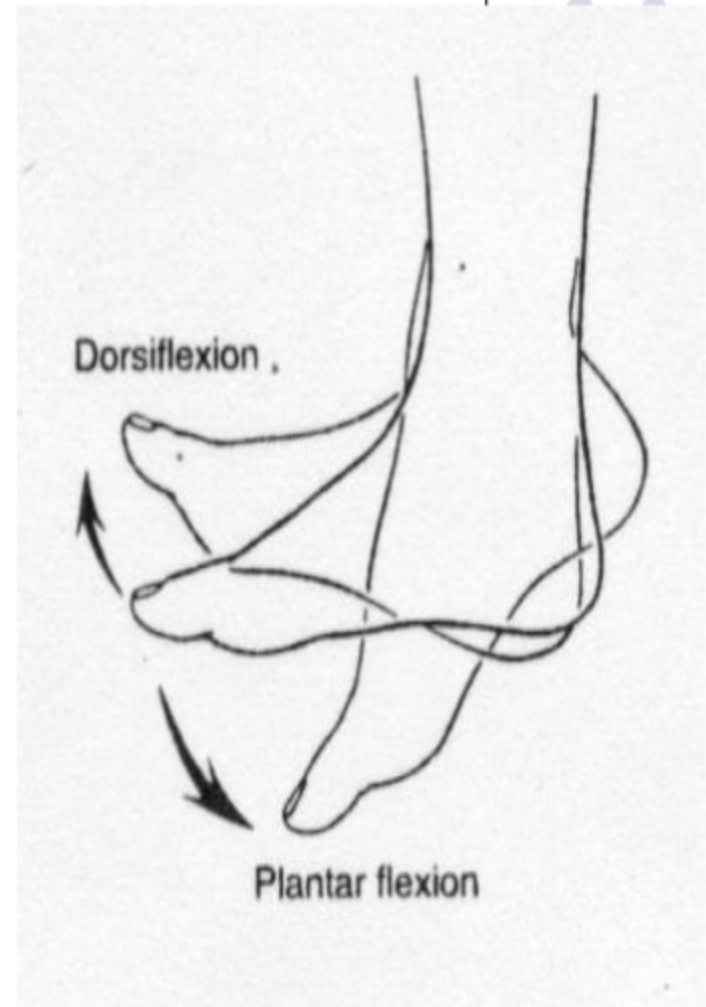


4. Plantarflexion/Dorsiflexion



□ Plantarflexion- Point toes/foot downward
“stand on tip toes”

□ Dorsiflexion- Point toes/foot upward
“foot off the gas pedal”



5. Inversion/ Eversion

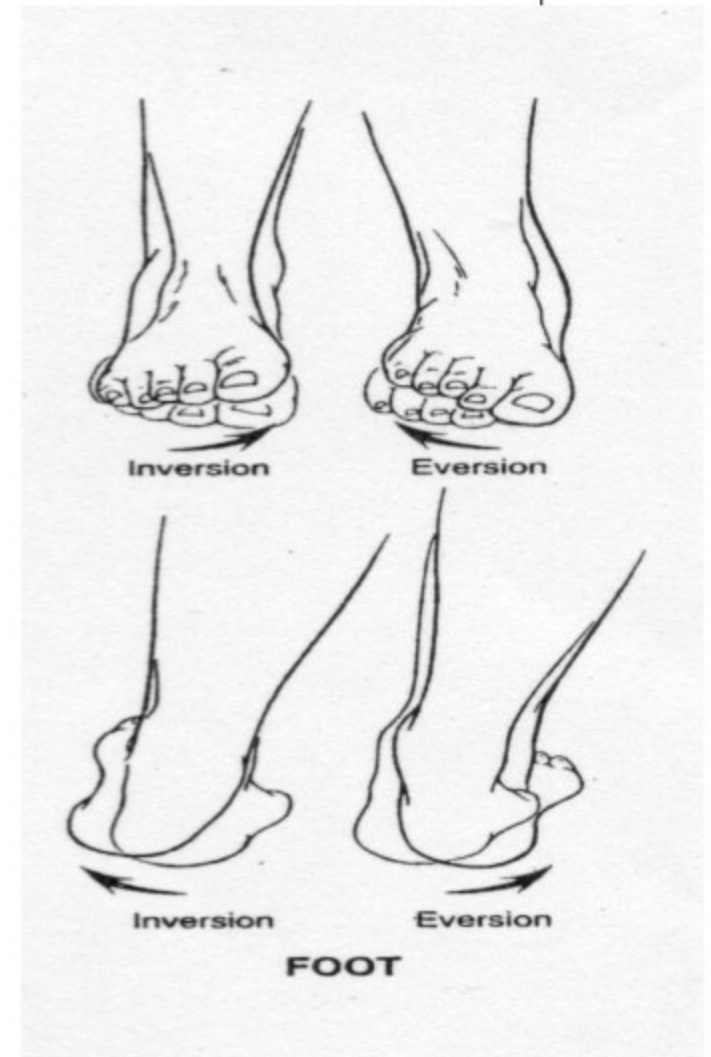


❖ Inversion

Movement of sole of foot inward

❖ Eversion-

Movement of sole of foot outward



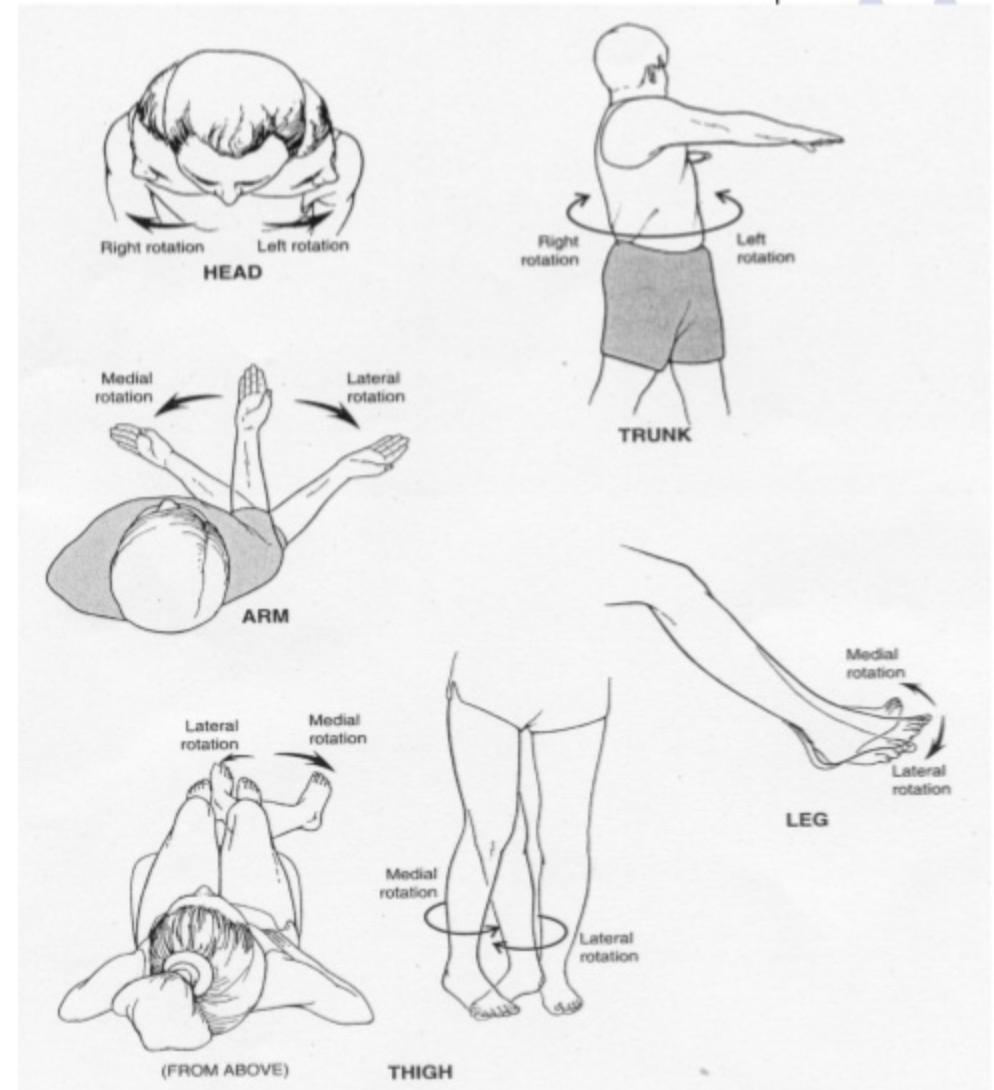
6. Medial/ Lateral

❖ Medial (Internal) Rotation

Anterior surface moves toward midline

❖ Lateral (External) Rotation

Anterior surface moves away from midline





7. Protraction/Retraction

❖ Protraction

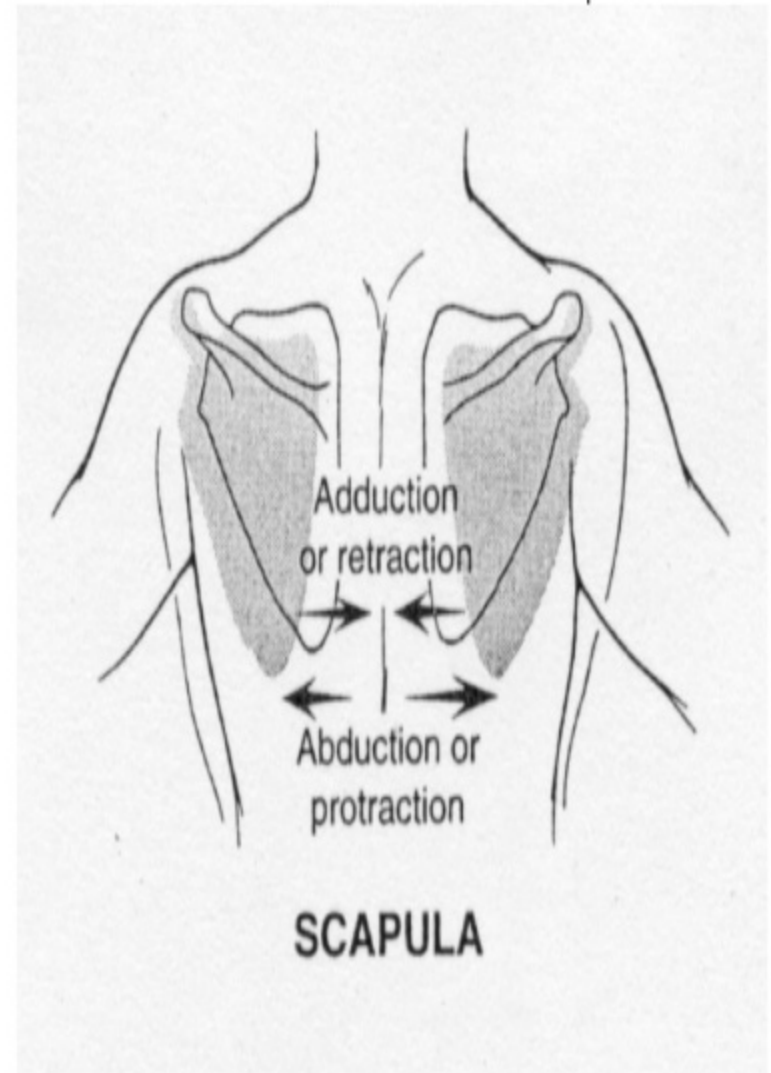
Moving forward on plane parallel to ground

“punch”

❖ Retraction

Moving backward on plane parallel to ground

“start lawn mower”



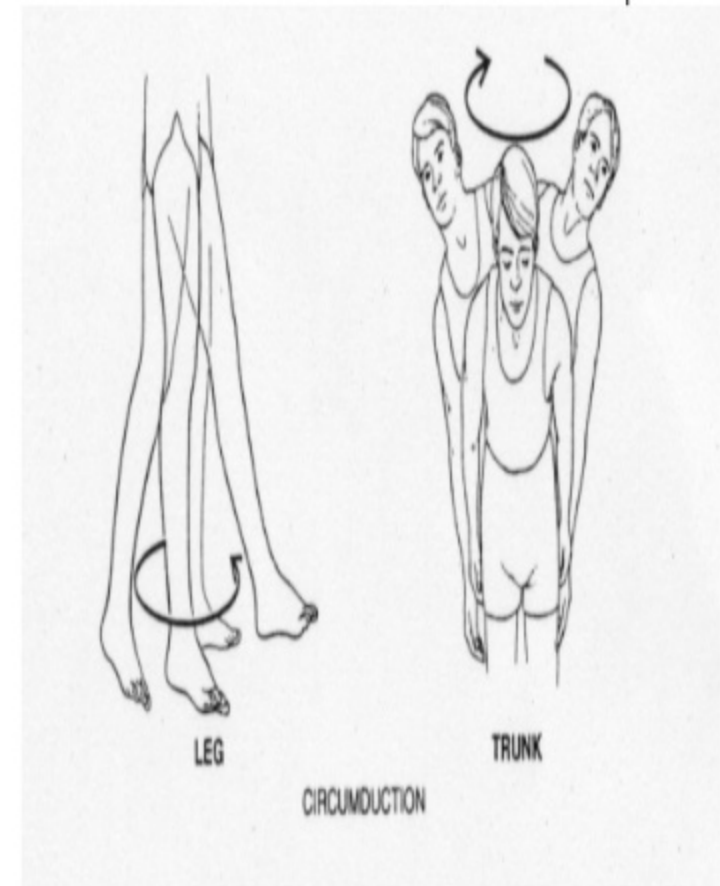
8. Circumduction



Circumduction

Latin meaning
“around”

Combination of
flexion/extension and
abduction/adduction





Thank You